Show-Ready Humanoids

Interactive and Customizable Robot Performers for Events and Entertainment

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THE PROBLEM

Event robots all do the same show.

Humanoid robots draw crowds at trade shows, brand launches and fashion events. But today they repeat the same pre-made dances at every event, their customization stops at stickers and logos, and their motion tools are built for one specific robot.

People watch once, then walk away.

OUR IDEA

Let the audience direct the robot.

We are building a platform that turns almost any humanoid robot into an interactive, customizable performer. Guests see their own moves performed by the robot, organizers launch any dance from a motion library, and brands dress the robot to match their look.

Every part comes from our previous and ongoing research at the ATR Lab.

At our booth today
Live demos of Mimic me and the Motion library. Customization is prepared per client.



PROOF OF CONCEPT
Hackathon-winning build: costume made in collaboration with a fashion team.

ENTERTAINMENT ROBOTICS

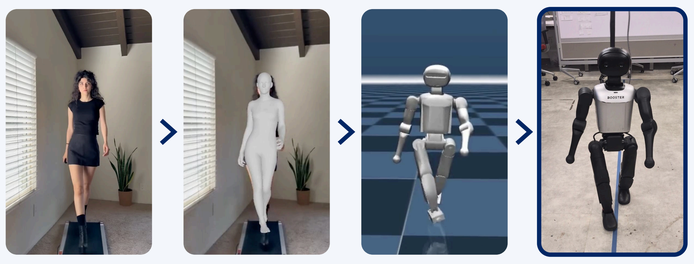
Three services, one platform

01 — Mimic me

LIVE AT OUR BOOTH

Immersive, personalized interaction

Every guest gets a personal moment: they show a move and the robot mimics it back, after a safety check in simulation.



Video-to-robot pipeline from our IROS 2026 work

< 1 min
from the guest's move to the robot

02 — Motion library

LIVE AT OUR BOOTH

Launch any dance in one click

A platform with a growing library of dances, poses and full routines. Pick a motion, press play, and the robot performs it.

One pretrained model runs the whole library. For a new routine we fine-tune it briefly instead of training from scratch.

97% of new movements reproduced (in simulation)

<1% of the original compute to move the model to a new robot

~10k human motion clips used to train the model

03 — Full customization

PRE-PREPARED

Prepared in advance for each client

The robot is adapted to a client's brand before the event: clothes, footwear, accessories and colors, designed around the robot so its sensors, cameras and joints stay clear.

Clothes and footwear change how a robot balances. Our ongoing research adapts the robot's motion to them, so it keeps performing in whatever it wears.

Clothes · Footwear · Accessories · Brand colors · Logos

The look and the moves change per event, not per robot.

TECHNOLOGY

Video capture: GVHMR · **Retargeting:** GMR, UMR · **Training:** NVIDIA Isaac Lab · **Pretrained model:** SONIC-K1 · **Testing:** MuJoCo simulation

BUSINESS MODEL

Robot-as-a-service

Clients pick one service or combine all three, for a single event or a season, in Northeast Ohio and beyond.

Events · Fashion · Trade shows · Retail · Sports · Entertainment

Let's run a pilot event

Looking for industry partners and venues

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